



AGRISCORE

Alternative Credit Scoring for Smallholder Farmers

KENYA / TANZANIA FOCUS

Capstone Project Proposal

GROUP 3 | May , 2026

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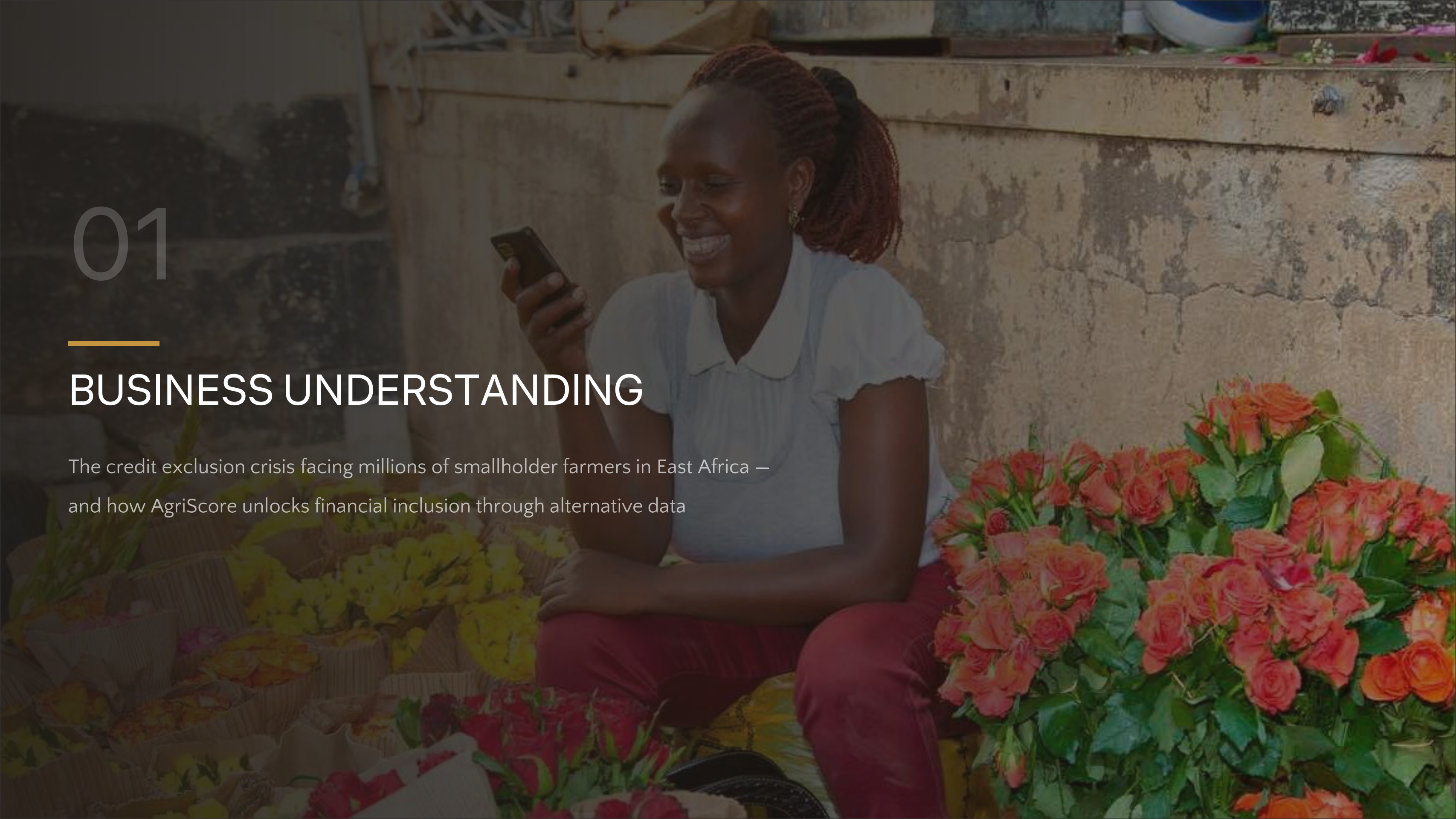
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Rigorous metrics: AUC-ROC >0.78, Gini >0.55, fairness audits

One-week MVP to full cloud-native API deployment

Streamlit dashboard + FastAPI + SMS/USSD fallback(future)



01

BUSINESS UNDERSTANDING

The credit exclusion crisis facing millions of smallholder farmers in East Africa —
and how AgriScore unlocks financial inclusion through alternative data

85%+ Mobile Money Penetration, Yet Farmers Remain Credit-Invisible

85%+

M-PESA penetration in Kenya — highest mobile money adoption globally

30-120%

Interest rates charged by informal lenders — exploitative and unsustainable

Millions

Of smallholder farmers (1-5 acres) excluded from formal credit systems

The Credit Exclusion Trap

Smallholder farmers lack traditional collateral, credit history, or formal banking records. Without access to credit, they cannot purchase quality seeds, fertilizer, or equipment — directly reducing yields and perpetuating poverty cycles.

The Paradox

Despite 85%+ M-PESA penetration generating rich digital footprints, financial institutions cannot leverage this data for credit decisions. The infrastructure exists; the scoring mechanism does not.



AgriScore Unlocks Inclusive Lending at the Point of Need

Machine Learning-Based Alternative Credit Scoring

AgriScore predicts loan repayment probability for farmers who lack traditional collateral by analyzing:

- Mobile money transactions (M-PESA proxy)
- Agricultural productivity data
- Socioeconomic indicators

Generates **instant, explainable credit scores** and risk tiers via API in under 3 seconds.



Instant Approvals — Score delivered in <3 seconds via API



Lower Defaults — Target 15-25% NPL reduction for lenders



Higher Yields — Farmers access quality inputs, improving productivity



Ethical by Design — Bias audits, SHAP explanations, fairness checks

Real-World Alignment

Aligns with national financial inclusion goals and proven pilots including **Apollo Agriculture** and **Pathways**. Builds on established signals: transaction consistency predicts repayment better than traditional collateral-based metrics.

Target Ecosystem: From Farmers to Regulators



Primary Beneficiaries

Smallholder Farmers

1-5 acre plots growing maize, beans, coffee, and other staple crops

Gain access to affordable credit for quality inputs, directly improving yields and household income



Direct Users

MFIs, SACCOs, Digital Lenders

Agri-input suppliers: MEA, Yara, and similar organizations

Reduce default risk, expand lending volumes, and reach previously unbanked farmer segments



Secondary Stakeholders

Regulators & Funders

CBK, KALRO, AGRA, IFC

Support national financial inclusion goals and climate-resilient agricultural policy with data-driven insights

Quantified Impact Targets

<3s

Instant credit score delivery via API

15-25%

Target NPL (non-performing loan) reduction

10k+

Farmers scalable in pilot phase

SHAP

Explainable AI with bias audits

02

DATA & MODELING

Multi-source data fusion, feature engineering, and a two-stage ML pipeline designed for accuracy, explainability, and fairness



Four Data Pillars Powering the Credit Score



Core

World Bank LSMS-ISA

Tanzania NPS Wave 5, Kenya equivalents, harmonized GitHub panels

Yields, inputs, sales, socioeconomic, financial access



Transactional

Synthetic Mobile Money

Kaggle MoMTSim, PaySim variants as M-PESA proxy with labels

RFM features, inflow/outflow patterns



Augmentation

SAGDA Library

Synthetic African agricultural data generation

Yields, soil properties, climate variables



Future

Partnerships & Satellite

KALRO (MOU), NDVI via Earth Engine, input suppliers, MFIs

Real data post-MVP via partnerships



Data Strategy: Free/public + synthetic data for MVP development. Real transactional and agricultural data via partnerships post-MVP. Estimated dataset: 10k-100k+ rows for initial modeling.

12+ Feature Categories Engineered for Repayment Prediction



Transactional

RFM (Recency, Frequency, Monetary)
Input purchase ratio
Inflow/outflow patterns



Agricultural

Yield stability across seasons
Fertilizer per hectare
Plot size, irrigation access



Socioeconomic

Household size & education
Asset ownership index
Dependency ratio



Derived

Seasonality indicators
Cashflow volatility index
SACCO membership proxy



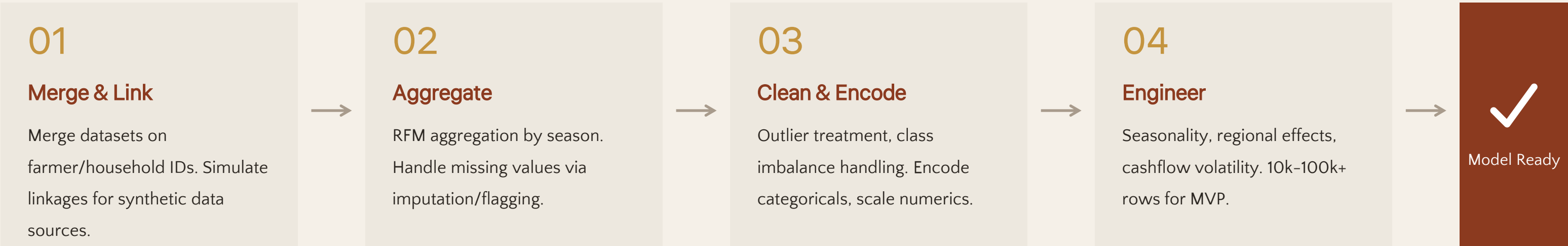
Key Insight

Transaction consistency — not collateral — is the strongest predictor of loan repayment. Farmers with regular, predictable mobile money flows default significantly less than those with volatile patterns, regardless of land ownership.

Prior Work & Foundation

Builds on established LSMS-ISA research, Apollo Agriculture and Pathways pilots, and alternative data credit models (RFM + agronomic features). East African fintech precedents include M-Shwari and M-KOPA, proving that non-traditional signals can effectively assess creditworthiness.

From Raw Data to Model-Ready Features



Key Challenges & Mitigations



EDA & Visualization

pandas profiling, seaborn for distributions and correlations. Yield vs. transaction pattern analysis. Risk heatmaps by region and crop type. SHAP summaries post-modeling for feature importance interpretation.

Two-Stage Model: Interpretability + Accuracy

STAGE 1 — BASELINE

Logistic Regression

Highly interpretable, regulatory-friendly baseline. Coefficients directly explain how each feature influences repayment probability — essential for compliance and farmer trust.



STAGE 2 — PRIMARY

XGBoost / LightGBM

Handles non-linearity, missing data, and feature interactions natively. Primary production model for maximum predictive accuracy. Hyperparameter tuning via Optuna/GridSearch.

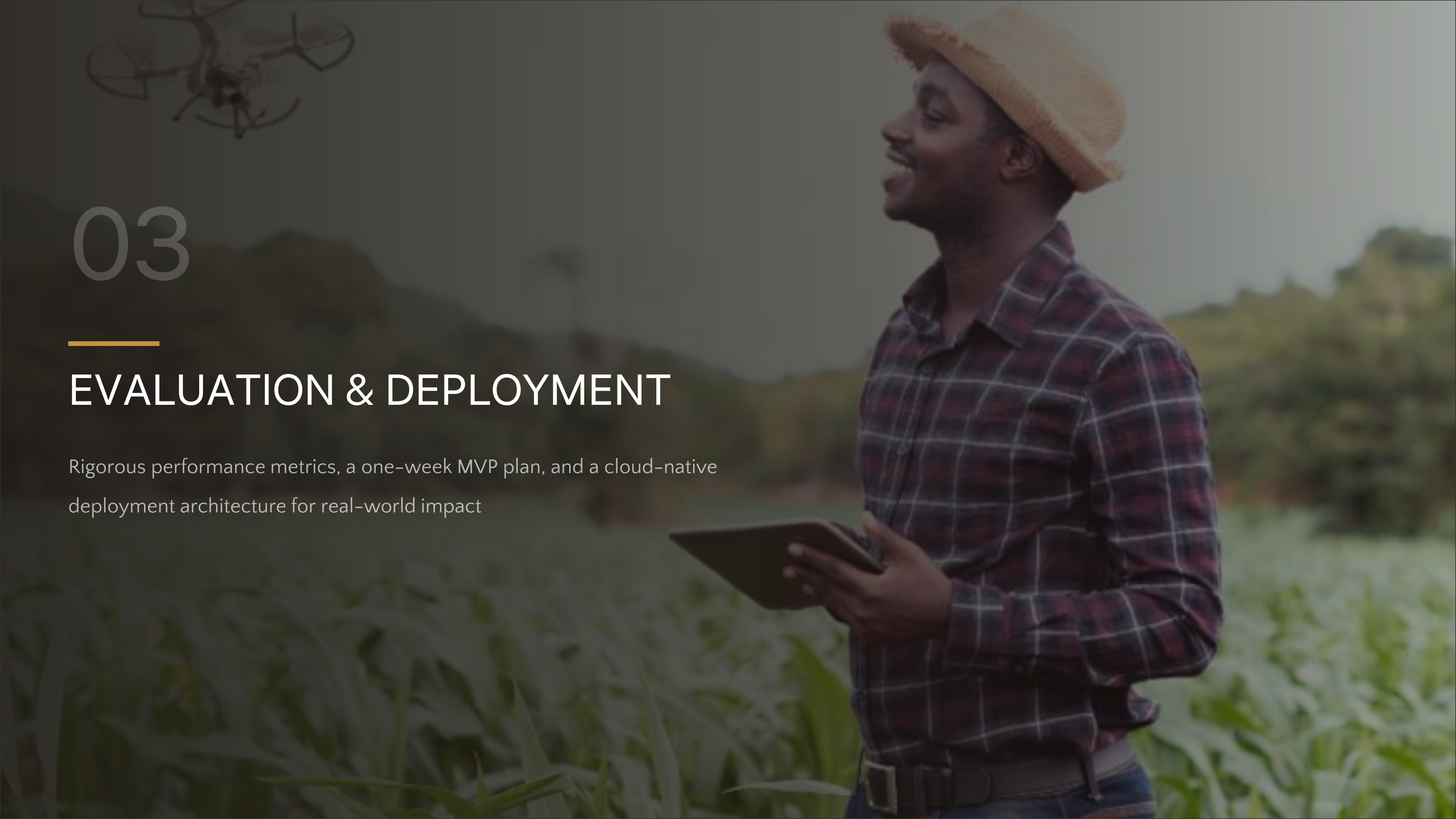


Explainability & Fairness

SHAP values explain every individual prediction. IBM AI Fairness 360 audits for demographic bias. Feature engineering is the critical bridge between raw data and model performance.

Target Variable

Supervised binary classification: **Repay vs. Default** — engineered from financial outcome labels. Class imbalance addressed through resampling and calibrated probability thresholds.



03

EVALUATION & DEPLOYMENT

Rigorous performance metrics, a one-week MVP plan, and a cloud-native deployment architecture for real-world impact

AUC-ROC >0.78 Target with Multi-Dimensional Evaluation

>0.78

AUC-ROC target for discrimination power

>0.55

Gini coefficient for model quality

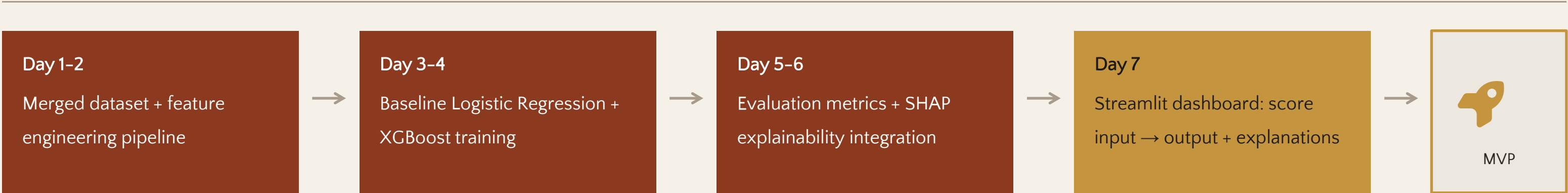
<0.05

Demographic parity fairness threshold

KS + PSI

Kolmogorov-Smirnov & Population Stability Index

MVP Timeline: 1 Week Sprint



Stretch Goals

Full API (FastAPI) · Satellite Integration (NDVI) · Bias Mitigation (AIF360) · Pilot Simulation · Model Card Documentation · Comparison Reports · Tanzania Expansion

Business impact: simulated default rate reduction, SHAP-driven insights for loan officers, and scalable risk-tier automation.

Cloud-Native Deployment with Rural Accessibility



User Interfaces

Interactive Streamlit dashboard (scores, SHAP explanations, risk tiers, PDF/CSV exports) + SMS/USSD fallback for rural access



API Layer

FastAPI REST endpoint: POST farmer data → returns score, risk tier, and loan ceiling recommendation



Infrastructure

Containerized (Docker), deployed to Render / Cloudflare / AWS / GCP Africa regions for low-latency regional access



Core Functionality

Instant scoring at PoS/loan app | Explainable decline reasons | Risk bands (Low/Med/High or Platinum-Gold) | Real-time monitoring dashboard

Python Ecosystem & CRISP-DM Governance

Category	Libraries & Tools
Data Processing	pandas, numpy, polars — high-performance tabular data manipulation
EDA & Prep	scikit-learn (preprocessing), feature-engine — transformations, encoding, scaling
Modeling	scikit-learn, xgboost, lightgbm, shap, imbalanced-learn — ML + explainability + imbalance handling
Viz & Dashboard	matplotlib, seaborn, plotly, Streamlit, Tableau — static to interactive visualizations

Algorithms & Environment

 **Core Algorithms**
Logistic Regression, XGBoost, LightGBM

 **Environment**
Local / Colab for MVP; AWS / GCP for scale

 **CRISP-DM + Ethics**
DPA 2019 compliance, consent, data minimization



AGRISCORE

Data-Driven Financial Inclusion for East Africa's Farmers

<3s

Instant Scoring

15-25%

NPL Reduction Target

10k+

Scalable Pilots

SHAP

Explainable AI

Built on proven signals, ethical design, and real-world partnerships.

Unlocking credit for millions who deserve it.

GROUP 3 | Capstone Project Proposal | May 5, 2026

AgriScore: Alternative Credit Scoring for Smallholder Farmers

THANKYOU :)